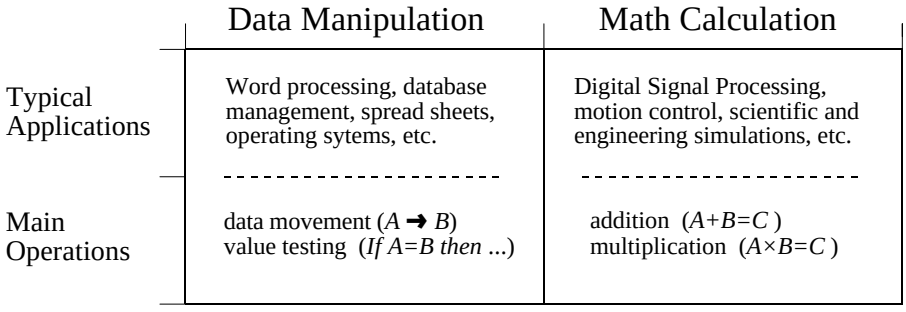

FIGURE 28-1
Data manipulation versus mathematical calculation. Digital computers are useful for two general tasks: *data manipulation* and *mathematical calculation*. Data manipulation is based on moving data and testing inequalities, while mathematical calculation uses multiplication and addition.

more important, there are *marketing* issues involved: development and manufacturing cost, competitive position, product lifetime, and so on. As a broad generalization, these factors have made traditional microprocessors, such as the Pentium®, primarily directed at data manipulation. Similarly, DSPs are designed to perform the mathematical calculations needed in Digital Signal Processing.

Figure 28-1 lists the most important differences between these two categories. Data manipulation involves storing and sorting information. For instance, consider a word processing program. The basic task is to store the information (typed in by the operator), organize the information (cut and paste, spell checking, page layout, etc.), and then retrieve the information (such as saving the document on a floppy disk or printing it with a laser printer). These tasks are accomplished by *moving* data from one location to another, and *testing* for inequalities ($A=B$, $A<B$, etc.). As an example, imagine sorting a list of words into alphabetical order. Each word is represented by an 8 bit number, the ASCII value of the first letter in the word. Alphabetizing involved rearranging the order of the words until the ASCII values continually increase from the beginning to the end of the list. This can be accomplished by repeating two steps over-and-over until the alphabetization is complete. First, test two adjacent entries for being in alphabetical order (IF A>B THEN ...). Second, if the two entries are not in alphabetical order, switch them so that they are (A⇄B). When this two step process is repeated many times on all adjacent pairs, the list will eventually become alphabetized.

As another example, consider how a document is printed from a word processor. The computer continually tests the input device (mouse or keyboard) for the binary code that indicates "print the document." When this code is detected, the program moves the data from the computer's memory to the printer. Here we have the same two basic operations: moving data and inequality testing. While mathematics is occasionally used in this type of